input.1p

Groff

```
import java.util.*;

public class Main {

    static Scanner sc = new Scanner(System.in);

    // Question class
    static class Question {
        String question;
        String[] options;
        int answer;
        String explanation;

        Question(String question, String[] options,
                 int answer, String explanation) {
            this.question = question;
            this.options = options;
            this.answer = answer;
            this.explanation = explanation;
        }
    }

    // Quiz class
    static class Quiz {
        String title;
        ArrayList<Question> questions = new ArrayList<>();

        Quiz(String title) {
            this.title = title;
        }

        void addQuestion(Question q) {
            questions.add(q);
        }
    }

    // Student result class
    static class Student {
        String name;
        String quizName;
        int score;
        int total;

        Student(String name, String quizName,
                int score, int total) {
            this.name = name;
            this.quizName = quizName;
            this.score = score;
            this.total = total;
        }
    }

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```
    static ArrayList<Quiz> quizzes = new ArrayList<>();
    static ArrayList<Student> leaderboard = new ArrayList<>();

    public static void main(String[] args) {

        createDefaultQuizzes();

        while (true) {
            System.out.println("\n================================");
            System.out.println("       SMART QUIZ APPLICATION");
            System.out.println("================================");
            System.out.println("1. Student Login");
            System.out.println("2. Admin Login");
            System.out.println("3. View Leaderboard");
            System.out.println("4. Exit");

            int choice = readInt("Enter your choice: ");

            switch (choice) {
                case 1:
                    studentLogin();
                    break;

                case 2:
                    adminLogin();
                    break;

                case 3:
                    showLeaderboard();
                    break;

                case 4:
                    System.out.println(
                        "Thank you for using Smart Quiz!"
                    );
                    sc.close();
                    return;

                default:
                    System.out.println(
                        "Invalid choice! Enter 1 to 4."
                    );
            }
        }
    }

    // Create sample quizzes
    static void createDefaultQuizzes() {

        Quiz java = new Quiz("Java Programming");

        java.addQuestion(new Question(
```

```
            "Which keyword is used to inherit a class?",
            new String[]{"implements", "extends", "inherits", "super"},
            2,
            "The extends keyword is used for class inheritance."
        ));

        java.addQuestion(new Question(
            "Which method is the entry point of Java?",
            new String[]{"start()", "run()", "main()", "init()"},
            3,
            "The main method is the standard entry point."
        ));

        java.addQuestion(new Question(
            "Which data type stores true or false?",
            new String[]{"int", "char", "boolean", "float"},
            3,
            "The boolean data type stores true or false."
        ));

        java.addQuestion(new Question(
            "Which keyword creates an object?",
            new String[]{"class", "new", "void", "static"},
            2,
            "The new keyword creates an object."
        ));

        java.addQuestion(new Question(
            "Which loop executes at least once?",
            new String[]{"for", "while", "do-while", "if"},
            3,
            "A do-while loop checks its condition after execution."
        ));

        Quiz python = new Quiz("Python Programming");

        python.addQuestion(new Question(
            "Which symbol starts a Python comment?",
            new String[]{"//", "#", "/*", "<!--"},
            2,
            "The # symbol starts a single-line Python comment."
        ));

        python.addQuestion(new Question(
            "Which function displays output in Python?",
            new String[]{"show()", "display()", "print()", "output()"},
            3,
            "The print() function displays output."
        ));

        python.addQuestion(new Question(
            "Which data structure stores key-value pairs?",
```

```
            new String[]{"List", "Tuple", "Dictionary", "Set"},
            3,
            "A dictionary stores key-value pairs."
        ));

        quizzes.add(java);
        quizzes.add(python);
    }

    // Student login
    static void studentLogin() {

        sc.nextLine();

        System.out.print("Enter your name: ");
        String name = sc.nextLine().trim();

        if (name.isEmpty()) {
            System.out.println("Name cannot be empty!");
            return;
        }

        System.out.println("\nWelcome, " + name + "!");

        if (quizzes.isEmpty()) {
            System.out.println("No quizzes available.");
            return;
        }

        System.out.println("\nAvailable Quizzes:");

        for (int i = 0; i < quizzes.size(); i++) {
            Quiz q = quizzes.get(i);

            System.out.println(
                (i + 1) + ". " + q.title +
                " (" + q.questions.size() + " questions)"
            );
        }

        int choice = readInt("Select a quiz: ");

        if (choice < 1 || choice > quizzes.size()) {
            System.out.println("Invalid quiz selection!");
            return;
        }

        Quiz selectedQuiz = quizzes.get(choice - 1);

        if (selectedQuiz.questions.isEmpty()) {
            System.out.println("This quiz has no questions yet.");
            return;
```

```
        }

        startQuiz(name, selectedQuiz);
    }

    // Start quiz and calculate score
    static void startQuiz(String name, Quiz quiz) {

        int score = 0;
        int total = quiz.questions.size();

        System.out.println("\n==================================");
        System.out.println("QUIZ: " + quiz.title);
        System.out.println("==================================");

        for (int i = 0; i < total; i++) {

            Question q = quiz.questions.get(i);

            System.out.println("\nQuestion " + (i + 1)
                               + " of " + total);
            System.out.println(q.question);

            for (int j = 0; j < q.options.length; j++) {
                System.out.println(
                    (j + 1) + ". " + q.options[j]
                );
            }

            int answer = readInt("Your answer (1-4): ");

            while (answer < 1 || answer > 4) {
                System.out.println(
                    "Invalid answer! Choose from 1 to 4."
                );
                answer = readInt("Your answer: ");
            }

            if (answer == q.answer) {
                score++;
                System.out.println("Correct answer!");
            } else {
                System.out.println("Incorrect answer!");
                System.out.println(
                    "Correct answer: " +
                    q.options[q.answer - 1]
                );
            }

            System.out.println(
                "Explanation: " + q.explanation
            );
```

```
261         }
262
263         showResult(name, quiz, score, total);
264     }
265
266     // Display final result
267     static void showResult(String name, Quiz quiz,
268                            int score, int total) {
269
270         double percentage =
271             total == 0 ? 0 : score * 100.0 / total;
272
273         System.out.println("\n==================================");
274         System.out.println("           QUIZ RESULT");
275         System.out.println("==================================");
276         System.out.println("Student Name: " + name);
277         System.out.println("Quiz Name: " + quiz.title);
278         System.out.println("Total Questions: " + total);
279         System.out.println("Correct Answers: " + score);
280         System.out.println("Wrong Answers: " + (total - score));
281
282         System.out.printf(
283             Locale.US, "Percentage: %.2f%%%n", percentage
284         );
285
286         if (percentage >= 90) {
287             System.out.println("Grade: A+");
288             System.out.println("Excellent performance!");
289         } else if (percentage >= 75) {
290             System.out.println("Grade: A");
291             System.out.println("Very good performance!");
292         } else if (percentage >= 60) {
293             System.out.println("Grade: B");
294             System.out.println("Good job! Keep practising.");
295         } else if (percentage >= 40) {
296             System.out.println("Grade: C");
297             System.out.println("Revise and practise more.");
298         } else {
299             System.out.println("Grade: D");
300             System.out.println("Keep learning and try again!");
301         }
302
303         leaderboard.add(
304             new Student(name, quiz.title, score, total)
305         );
306
307         System.out.println("==================================");
308     }
309
310     // Admin login
311     static void adminLogin() {
312
```

```
313         sc.nextLine();
314
315         System.out.print("Enter admin username: ");
316         String username = sc.nextLine();
317
318         System.out.print("Enter admin password: ");
319         String password = sc.nextLine();
320
321         if (!username.equals("admin")
322                 || !password.equals("admin123")) {
323             System.out.println("Invalid admin credentials!");
324             return;
325         }
326
327         System.out.println("\nAdmin login successful!");
328
329         while (true) {
330             System.out.println("\n========== ADMIN DASHBOARD ==========");
331             System.out.println("1. Create Quiz");
332             System.out.println("2. Add Question");
333             System.out.println("3. View All Quizzes");
334             System.out.println("4. View Leaderboard");
335             System.out.println("5. Logout");
336
337             int choice = readInt("Enter your choice: ");
338
339             switch (choice) {
340                 case 1:
341                     createQuiz();
342                     break;
343
344                 case 2:
345                     addQuestion();
346                     break;
347
348                 case 3:
349                     viewQuizzes();
350                     break;
351
352                 case 4:
353                     showLeaderboard();
354                     break;
355
356                 case 5:
357                     System.out.println("Admin logged out.");
358                     return;
359
360                 default:
361                     System.out.println("Invalid choice!");
362             }
363         }
364     }
```

```
365
366     // Create a new quiz
367     static void createQuiz() {
368
369         sc.nextLine();
370
371         System.out.print("Enter quiz title: ");
372         String title = sc.nextLine().trim();
373
374         if (title.isEmpty()) {
375             System.out.println("Quiz title cannot be empty!");
376             return;
377         }
378
379         quizzes.add(new Quiz(title));
380
381         System.out.println(
382             "Quiz '" + title + "' created successfully!"
383         );
384     }
385
386     // Add a question to a quiz
387     static void addQuestion() {
388
389         if (quizzes.isEmpty()) {
390             System.out.println("Create a quiz first!");
391             return;
392         }
393
394         viewQuizzes();
395
396         int index = readInt("Select quiz number: ");
397
398         if (index < 1 || index > quizzes.size()) {
399             System.out.println("Invalid quiz number!");
400             return;
401         }
402
403         sc.nextLine();
404
405         System.out.print("Enter question: ");
406         String question = sc.nextLine().trim();
407
408         if (question.isEmpty()) {
409             System.out.println("Question cannot be empty!");
410             return;
411         }
412
413         String[] options = new String[4];
414
415         for (int i = 0; i < 4; i++) {
416             System.out.print("Enter option " + (i + 1) + ": ");
```

```
            options[i] = sc.nextLine().trim();

            if (options[i].isEmpty()) {
                System.out.println("Option cannot be empty!");
                return;
            }
        }

        int answer = readInt("Enter correct option (1-4): ");

        while (answer < 1 || answer > 4) {
            System.out.println("Choose a number from 1 to 4.");
            answer = readInt("Enter correct option: ");
        }

        sc.nextLine();

        System.out.print("Enter answer explanation: ");
        String explanation = sc.nextLine().trim();

        quizzes.get(index - 1).addQuestion(
            new Question(question, options, answer, explanation)
        );

        System.out.println("Question added successfully!");
    }

    // Display all quizzes
    static void viewQuizzes() {

        System.out.println("\n========== AVAILABLE QUIZZES ==========");

        if (quizzes.isEmpty()) {
            System.out.println("No quizzes available.");
            return;
        }

        for (int i = 0; i < quizzes.size(); i++) {
            Quiz q = quizzes.get(i);

            System.out.println(
                (i + 1) + ". " + q.title +
                " | Questions: " + q.questions.size()
            );
        }
    }

    // Display leaderboard
    static void showLeaderboard() {

        if (leaderboard.isEmpty()) {
            System.out.println("\nNo quiz results available yet.");
```

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```
469             return;
470         }
471
472         ArrayList<Student> sorted =
473             new ArrayList<>(leaderboard);
474
475         sorted.sort((a, b) -> {
476             int percentageA =
477                 a.score * 100 / a.total;
478             int percentageB =
479                 b.score * 100 / b.total;
480
481             return Integer.compare(percentageB, percentageA);
482         });
483
484         System.out.println("\n========== LEADERBOARD ==========");
485
486         for (int i = 0; i < sorted.size(); i++) {
487             Student s = sorted.get(i);
488
489             double percentage =
490                 s.total == 0 ? 0 :
491                 s.score * 100.0 / s.total;
492
493             System.out.printf(
494                 Locale.US,
495                 "%d. %s | %s | Score: %d/%d | %.2f%%%n",
496                 i + 1, s.name, s.quizName,
497                 s.score, s.total, percentage
498             );
499         }
500     }
501
502     // Safe integer input
503     static int readInt(String message) {
504
505         while (true) {
506             System.out.print(message);
507
508             if (sc.hasNextInt()) {
509                 return sc.nextInt();
510             }
511
512             System.out.println("Invalid input! Enter a number.");
513             sc.next();
514         }
515     }
516 }
517
```